

Ex 1

31

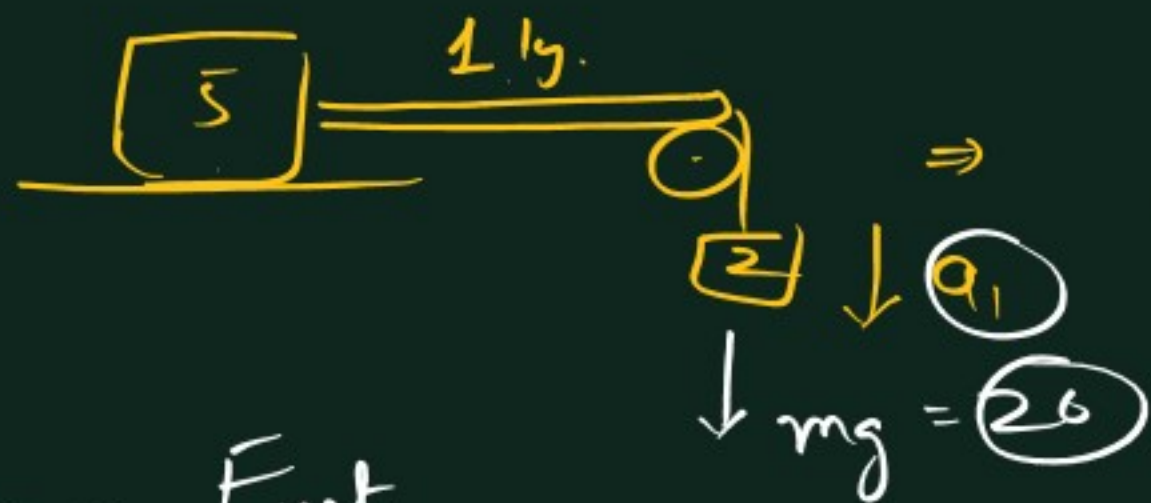


$\equiv$

Same a

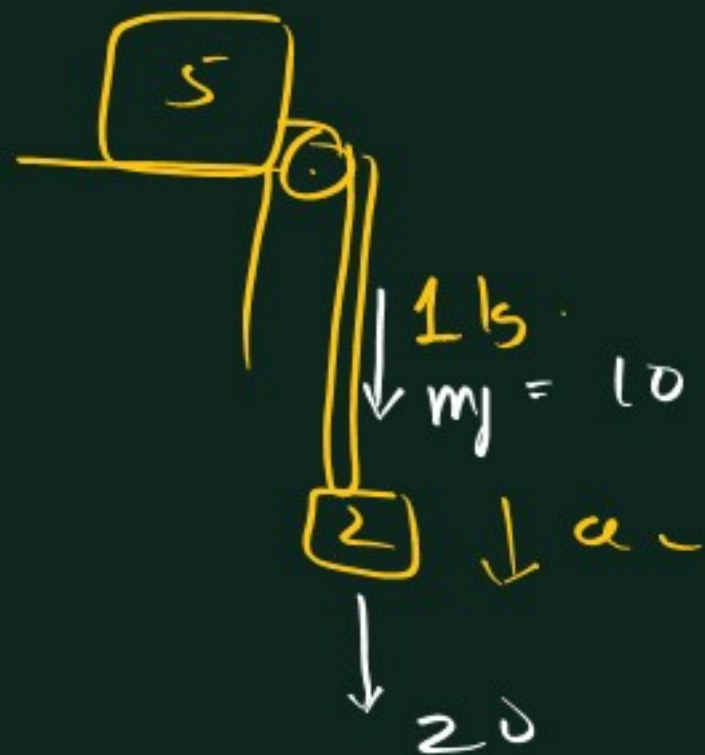


(25)



$$a_1 = \frac{F_{\text{net}}}{\text{Total mass}}$$

$$= \frac{20}{8 \text{ kg}} = \frac{5}{2}$$



$$a = \frac{10 + 20}{8} = \frac{30}{8} = \frac{15}{4}$$

Kinematics → Study of motion
(a, v, s etc)

Types of motion

i) 1D, 2D, 3D

Rectilinear motion

In Plane

• Projectile

• Circular 

Fly, Plane

② i) uniform motion

→ constant velocity ($a=0$)

Distance = Speed × time

ii) uniformly accelerated motion

→ $a = \text{constant}$

→ Equations of motion

iii) Non-uniformly accelerated motion

→ $a = \text{changes with time}$

↓
Calculus

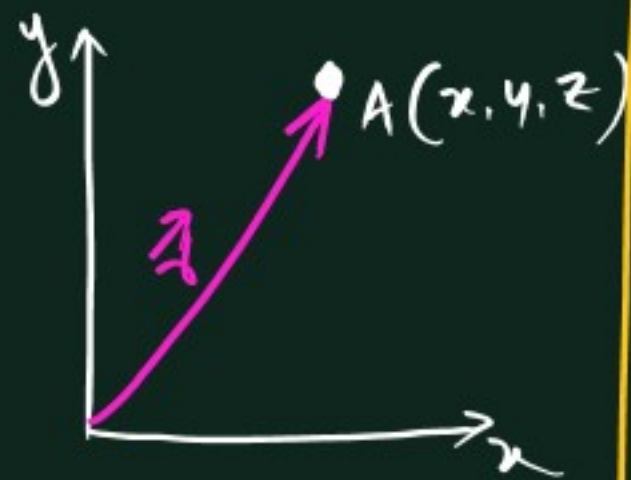
$$a = 2 \text{ m/s}^2 \\ = \frac{(2 \text{ m/s})}{s}$$

$$\begin{array}{cc} 5 \text{ m/s} & 5+2=7 \\ \text{---} \rightarrow & \text{---} \rightarrow \\ t=0 & \end{array}$$

$$\begin{array}{cc} \Downarrow & \\ 7+2=9 & 9+2=11 \\ \text{---} \rightarrow & \text{---} \rightarrow \end{array}$$

1) Position vector ($\vec{r}$)

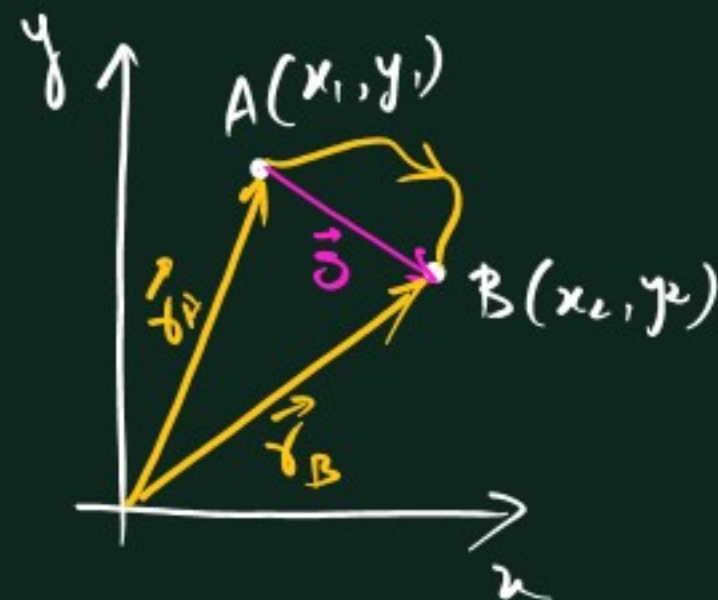
$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$



Ex: $A(3, -2, 6)$

$$\vec{r}_A = 3\hat{i} - 2\hat{j} + 6\hat{k}$$

2) Displacement ($\vec{S}$)



$$\vec{r}_A + \vec{S} = \vec{r}_B$$

$$\Rightarrow \boxed{\vec{S} = \vec{r}_B - \vec{r}_A}$$

$$\Rightarrow \vec{S} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j}$$

$$|\vec{S}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Q

$A(1, 3, -2)$
 $B(4, 1, 6)$

Object moves A to B. Find

1) Disp. vector

2) Magnitude of disp.

$$\vec{S} = \vec{r}_2 - \vec{r}_1$$

$$= (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$$

$$= (4 - 1)\hat{i} + (1 - 3)\hat{j} + (6 - (-2))\hat{k}$$

$$\boxed{\vec{S} = 3\hat{i} - 2\hat{j} + 8\hat{k}}$$

$$|\vec{S}| = \sqrt{3^2 + 2^2 + 8^2}$$

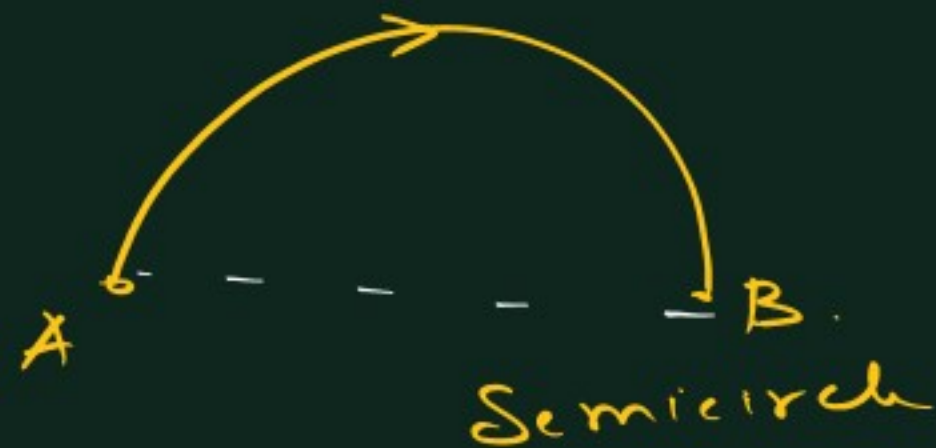
$$= \underline{\underline{\sqrt{77} \text{ m}}}$$

$$\text{Distance} \geq |\text{Displacement}|$$

Q When Distance = |Disp. |

A 1D motion $\rightarrow$ without changing direction.

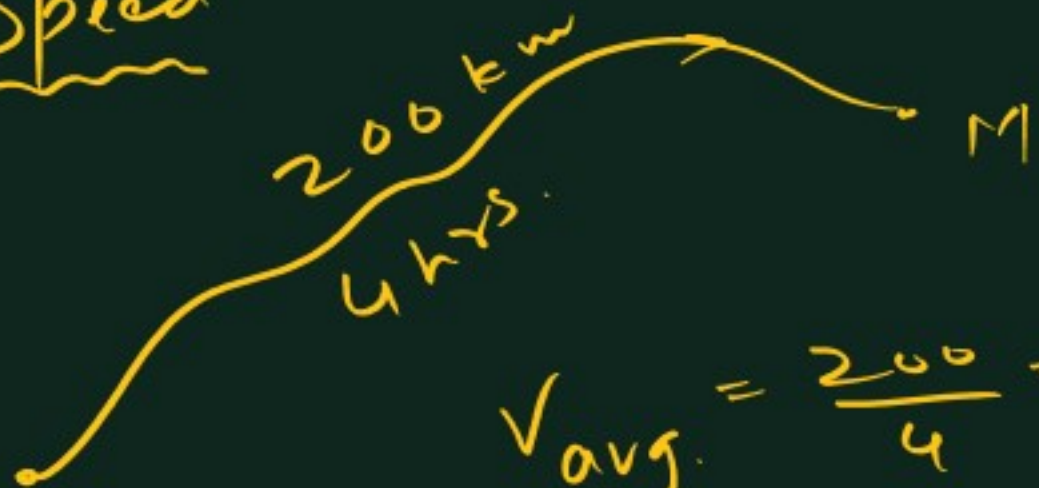
Q



Find ratio of
Distance : |Disp| from
A to B.

$$A = \frac{\text{Dist.}}{\text{Disp.}} = \frac{(\pi R)}{2R} = \left(\frac{\pi}{2} \right)$$

Speed



$$V_{avg} = \frac{200}{4} = 50 \text{ km/hr.}$$

~~Q~~ Avg. speed = $\frac{\text{Total Distance}}{\text{total time}}$

→ Instantaneous speed = Speed at any instant
(Speedometer)

$$\left(V_{inst.} = \frac{ds}{dt} \right)$$

$s = \text{distance}$

Velocity

$$\text{Avg. velocity, } \vec{V}_{avg} = \frac{\text{Total disp.}}{\text{time}}$$

Inst. velocity = velocity at instant

$$\boxed{\text{Avg. speed} \geq |\text{Avg. velocity}|}$$

$$\boxed{\text{Inst. speed} = |\text{Inst. velocity}|}$$

Q1.



Person completes 1 circle
in 30 seconds.

Find 1) Avg. speed
2) Avg. velocity.

Ans.

$$\text{Distance} = 2\pi r$$

$$= 4\pi$$

$$\text{Avg. speed} = \frac{4\pi}{30} \text{ m/s}$$

$$= \left(\frac{2\pi}{15}\right) \text{ m/s}$$

$$\text{Avg. vel.} = 0$$

Q2.



Find avg. speed from A to C.

Ans.

$$\text{Avg. speed} = \frac{AB + BC}{10 + 20}$$

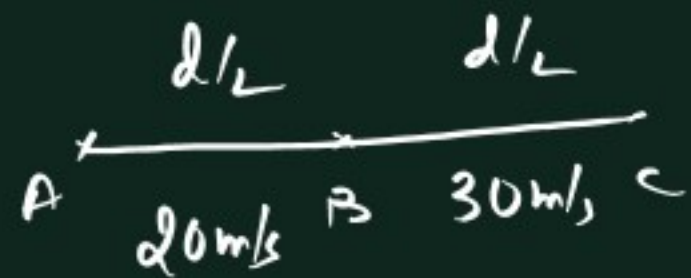
$$AB = v \times t = 20 \times 10 = 200$$

$$BC = v \times t = 40 \times 20 = 800$$

$$\text{Avg. speed} = \frac{200 + 800}{30}$$

$$= \frac{1000}{3} = \underline{\underline{333.3 \text{ m/s}}}$$

3) let total distance = d (or) let total distance = 120m



$$\text{Avg. speed} = \frac{\text{Total distance}}{\text{total time}}$$

$$= \frac{d}{\left(\frac{d/2}{20}\right) + \left(\frac{d/2}{30}\right)}$$

$$\left(t = \frac{\text{Distance}}{\text{Speed}}\right)$$

$$\text{Avg. speed} = 24 \text{ m/s}$$

$$\begin{array}{cc} \frac{60}{20 \text{ m/s}} & \frac{60}{30 \text{ m/s}} \\ \downarrow & \downarrow \\ t_1 = 3 \text{ s} & t_2 = 2 \text{ s} \end{array} \quad \left| \begin{array}{l} V_{\text{avg}} = \frac{120}{3+2} \\ = 24 \end{array} \right.$$



AB $\rightarrow \frac{1}{3}$ rd of total time

BC $\rightarrow \frac{2}{3}$ rd

Avg. speed A to C.

$$A_{\text{avg}} = \frac{30 \text{ m/s} \times \frac{t}{3} + 60 \text{ m/s} \times \frac{2t}{3}}{t}$$

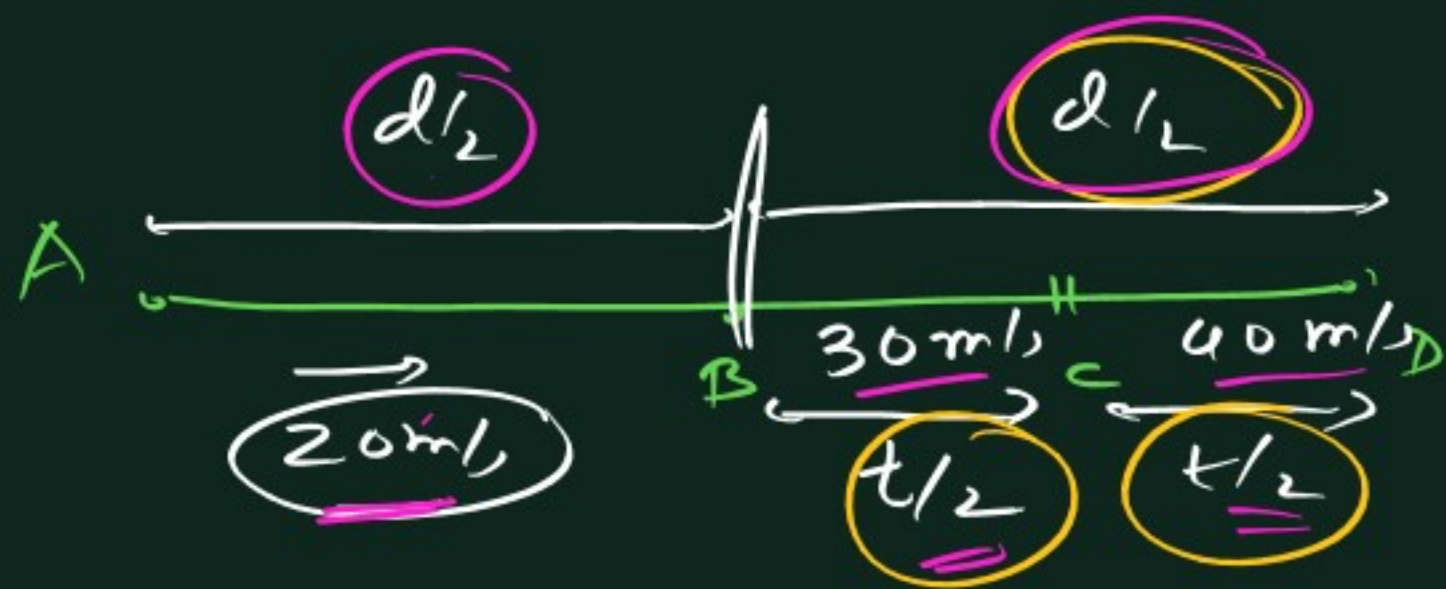
$$V_{\text{avg}} = \frac{30 \times \frac{t}{3} + 60 \times \frac{2t}{3}}{t}$$

$$= 50 \text{ m/s}$$

let total time = 6s

$$\frac{30 \text{ m/s} \times 2 \text{ s} + 60 \text{ m/s} \times 4 \text{ s}}{2+4}$$

$$V_{\text{avg}} = \frac{30 \times 2 + 60 \times 4}{2+4} = 50$$



Avg. speed = $\frac{d}{\left(\frac{d/2}{20}\right) + \frac{t}{2} + \frac{t}{2}}$

= $\frac{d}{d/40 + t}$ — (1)

Express t in terms of d

B to D.

$$\frac{d}{2} = \frac{t}{2} \times 30 + \frac{t}{2} \times 40$$

$$\Rightarrow d = 70t$$

$$V_{\text{avg}} = \frac{70t}{\frac{70t}{40} + t}$$

$$= \frac{7(70)}{7\left(\frac{7}{4} + 1\right)}$$

$$= \frac{280}{11} \text{ m/s} = 25.45 \text{ m/s}$$

$$a = 3 \text{ m/s}^2 \text{ or } \left(\frac{3 \text{ m/s}}{\text{s}} \right)$$

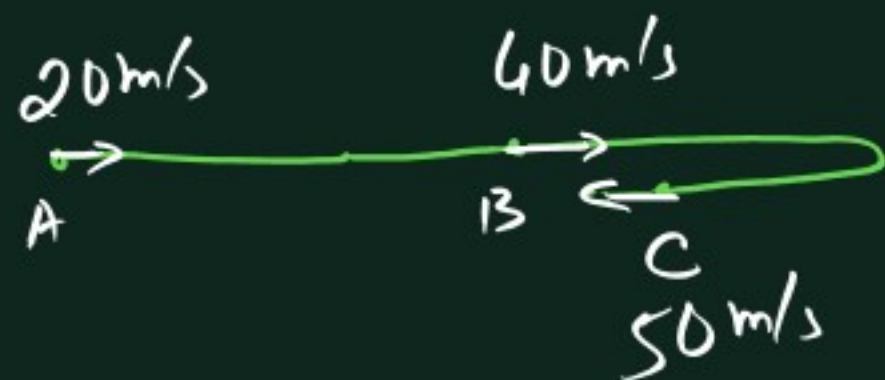
Avg. acceleration, $\vec{a}_{\text{avg}} = \frac{\text{Change in velocity}}{\text{time}} = \frac{\vec{v}_2 - \vec{v}_1}{t}$ (subtraction of vector)

$$\left(a = \frac{dv}{dt} \right)$$

$$\underline{V_{\text{avg}} = \frac{2v_1 v_2}{v_1 + v_2}}$$

$$V_{\text{avg}} = \frac{v_1 + v_2}{2}$$

Q 1D motion.



A to B $\Rightarrow$ 5 sec.

B to C $\Rightarrow$ 10 sec.

Find avg. acceleration (magnitude)

from 1) A to B

2) B to C.

A) 12

B) 4

~~C) 1~~

D) 9

E) 45

A to B

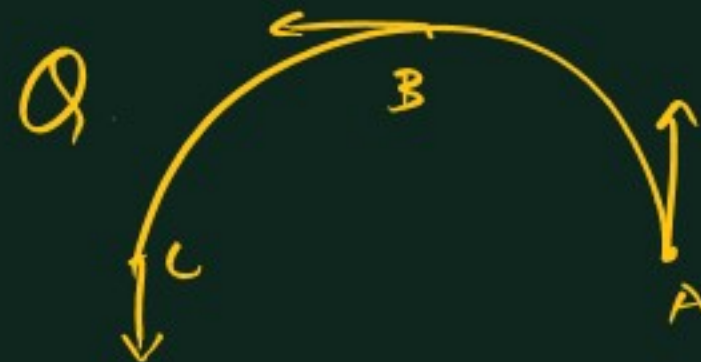
$$a_{\text{avg}} = \frac{V_B - V_A}{t}$$
$$= \frac{40 - 20}{5}$$
$$= 4 \text{ m/s}^2$$

B to C

$$a_{\text{avg}} = \frac{V_C - V_B}{t}$$
$$= \frac{(-50) - (+40)}{10}$$

$$= -9 \text{ m/s}^2$$

$$|a_{\text{avg}}| = 9 \text{ m/s}^2$$



A to B = 4 sec.

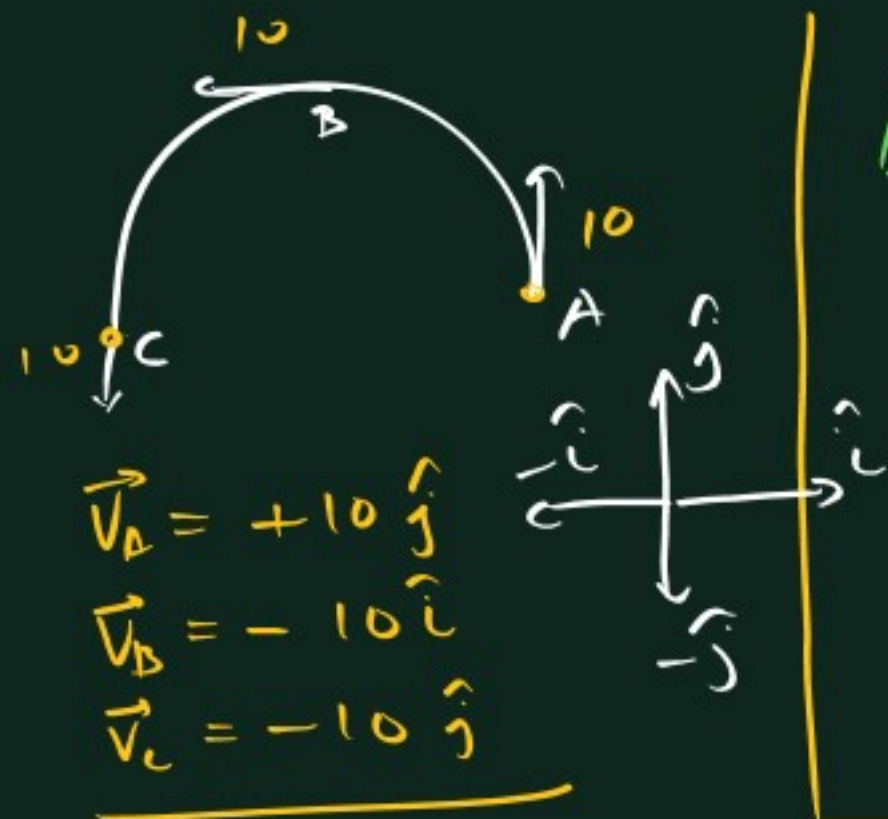
B to C = 4 sec.

Object moves at
Speed of 10 m/s.

Find avg. a from

1) A to C

2) A to B.



A to C

$$\bar{a}_{av} = \frac{\vec{V}_C - \vec{V}_A}{t} = \frac{(-10 \hat{j}) - (10 \hat{j})}{8}$$

$$= -\frac{20}{8} \hat{j} = \boxed{-2.5 \hat{j}}$$

$$|\bar{a}_{av}| = 2.5 \text{ m/s}^2$$

A to B

$$\bar{a}_{av} = \frac{\vec{V}_B - \vec{V}_A}{t}$$

$$= \frac{(-10 \hat{i}) - (10 \hat{j})}{4}$$

$$\bar{a}_{av} = \frac{-10 \hat{i} - 10 \hat{j}}{4}$$

$$|\bar{a}_{av}| = \sqrt{\left(\frac{10}{4}\right)^2 + \left(\frac{10}{4}\right)^2}$$

$$= \frac{10}{4} \sqrt{2}$$

$$= \boxed{\frac{5\sqrt{2}}{2}}$$

A to C

$$\vec{A} = x \hat{i} + y \hat{j}$$

$$|\vec{A}| = \sqrt{x^2 + y^2}$$

$V_A = +10, V_C = -10$

$$\bar{a}_{av} = \frac{V_C - V_A}{t}$$

$$= \frac{(-10) - (10)}{8}$$

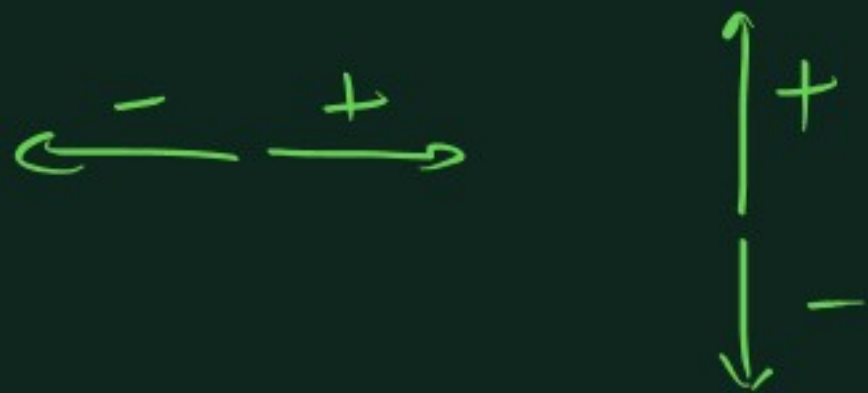
$$= -\frac{20}{8} = -2.5$$

ii) $\bar{a}_{av} = \frac{|\vec{V}_B - \vec{V}_A|}{t}$

$$= \frac{\sqrt{10^2 + 10^2} - 2 \times 10 \cos 90^\circ}{4}$$

$$= \frac{\sqrt{200}}{4} = \boxed{\frac{10\sqrt{2}}{4}}$$

Equations of motion.



$$1) v = u + at$$

$$2) s = ut + \frac{1}{2}at^2$$

$$3) v^2 = u^2 + 2as$$

v = final velocity

u = initial "

s = Displacement

a = acc.

t = time

4)

$$\text{Avg. } v = \frac{\text{Disp.}}{\text{time}}$$

$$\text{Avg. velocity } v_{\text{avg}} = \left(\frac{u+v}{2} \right)$$

$$5) s = v_{\text{avg}} \times t = \left(\frac{u+v}{2} \right) t$$

6) Disp. in n^{th} second

$$S_{n^{\text{th}}} = u + \frac{a}{2}(2n-1)$$

Q1.

$$\begin{aligned} u &= 0 \\ \longrightarrow \\ a &= 10 \text{ m/s}^2 \end{aligned}$$

Find 1) speed after 4 sec

2) Distance in 0 to 4s

3) If initial position is at $x = 10 \text{ m}$, find position at $t = \underline{4 \text{ s}}$.

$$\left. \begin{aligned} u &= 0 \\ a &= 10 \\ t &= 4 \text{ s} \end{aligned} \right\}$$

$$1) v = u + at$$

$$\Rightarrow v = 0 + 10 \times 4 \\ = \underline{40 \text{ m/s}}$$

$$\begin{aligned} 2) S &= ut + \frac{1}{2} at^2 \\ &= 0 + \frac{1}{2} \times 10 \times 4^2 \\ &= 80 \text{ m} \end{aligned}$$

3)

$$S = V_{\text{avg}} \times t = \left(\frac{0+40}{2} \right) \times 4 = \underline{80 \text{ m}}$$

Disp. = change in position

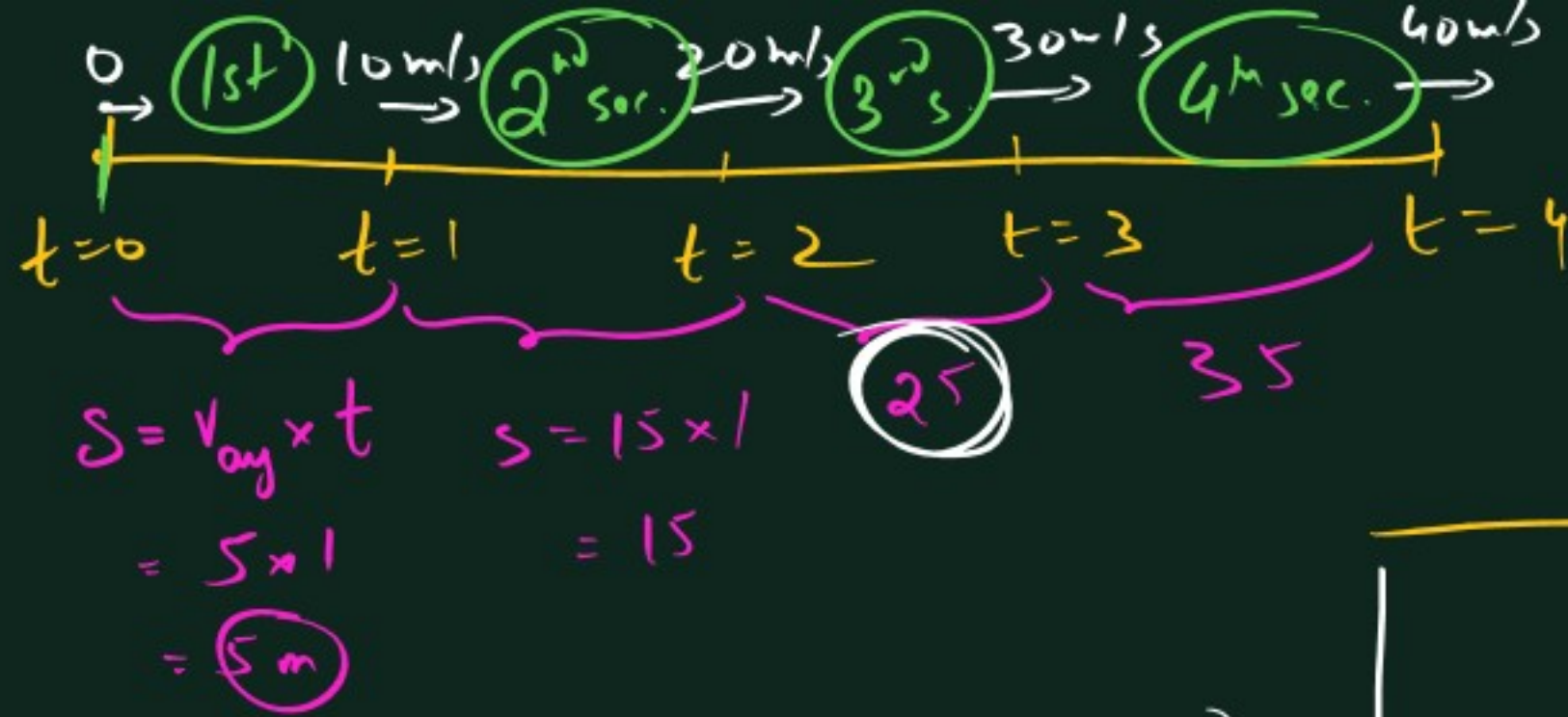
$$\text{or } S = x_2 - x_1$$

Here

$$S = 80 \text{ m}$$

$$\Rightarrow x_2 - \cancel{x_1} = 80$$

$$\Rightarrow \underline{x_2 = 90 \text{ m}}$$



Q Find disp. in 1st sec. & 3rd sec.
 $u=0, a=10$

$$S_{1st} = u + \frac{a}{2}(2n-1)$$

$$= 0 + \frac{10}{2}(2 \times 1 - 1) = 5 \text{ m}$$

$$S_{3rd} = u + \frac{a}{2}(2 \times 3 - 1)$$

$$= 0 + 5(5)$$

$$= 25 \text{ m}$$

$S_{nth} \Rightarrow$ Disp. between
 $t = (n-1)$ to $t = n$

Q Prove disp. in n^{th} sec.
 is $S_{nth} = u + \frac{a}{2}(2n-1)$

Diagram showing a horizontal line with a point at $t=0$ and a point at $t=n-1$. A green bracket indicates the displacement between $t=n-1$ and $t=n$.

$$S_{nth} = (\text{Disp. in } n \text{ sec.}) - (\text{Disp. in } (n-1) \text{ sec.})$$

$$S_{nth} = (S \text{ in } n) - (S \text{ in } (n-1))$$

$$= \left(u + \frac{1}{2}at_n \right) - \left(u + \frac{1}{2}a(n-1) \right)$$

$$= u(n - (n-1)) + \frac{1}{2}a(n^2 - (n-1)^2)$$

$$S_{nth} = u + \frac{a}{2}(2n-1)$$

Q $u = 20 \text{ m/s}$
 $t = 0 \rightarrow$
 $a = 8 \text{ m/s}^2$

Find out its displacement
 from $t = 2.5$ to $t = 3.5$
 seconds.

S_{nth} $n = 3.5$

$$\begin{aligned} S_{nth} &= u + \frac{a}{2}(2n-1) \\ &= 20 + \frac{8}{2}(2 \times 3.5 - 1) \\ &= 20 + 4 \times 6 \\ &= \underline{\underline{44 \text{ m}}} \end{aligned}$$

$u = 0$

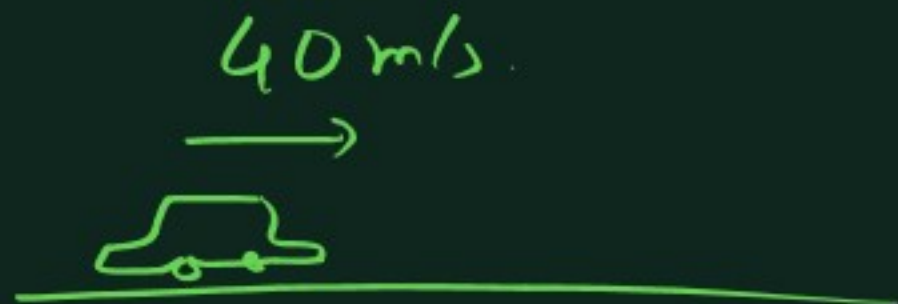
$$\begin{aligned} S_{1st} &= 0 + \frac{a}{2}(2 \times 1 - 1) \\ &= \frac{a}{2} \end{aligned}$$

$$\begin{aligned} S_{2nd} &= 0 + \frac{a}{2}(2 \times 2 - 1) \\ &= 3\left(\frac{a}{2}\right) \end{aligned}$$

$$\begin{aligned} S_{3rd} &= 0 + \frac{a}{2}(2 \times 3 - 1) \\ &= 5\left(\frac{a}{2}\right) \end{aligned}$$

Ratio = 1 : 3 : 5 : _

Q.



$$u = 40$$

$$a = -5$$

$$\underline{v = 0}$$

$$v = u + at$$

$$\Rightarrow 0 = 40 - 5t$$

$$\Rightarrow \underline{t = 8s.}$$

$$\begin{aligned} S &= ut + \frac{1}{2}at^2 \\ &= 40 \times 8 - \frac{5}{2} \times 8^2 \\ &= 160 \end{aligned}$$

(68)

$$\begin{aligned} S &= v_{av} \times t \\ &= 20 \times 8 \\ &= \underline{\underline{160m}} \end{aligned}$$

Q. 40 m/s
 $\longrightarrow$

$\longleftarrow$
 $a = 10 \text{ m/s}^2$

$$u = +40$$

$$a = -10$$

$$t = 3 \text{ s}$$

$$v = u + at$$

$$= 40 - 10 \times 3$$

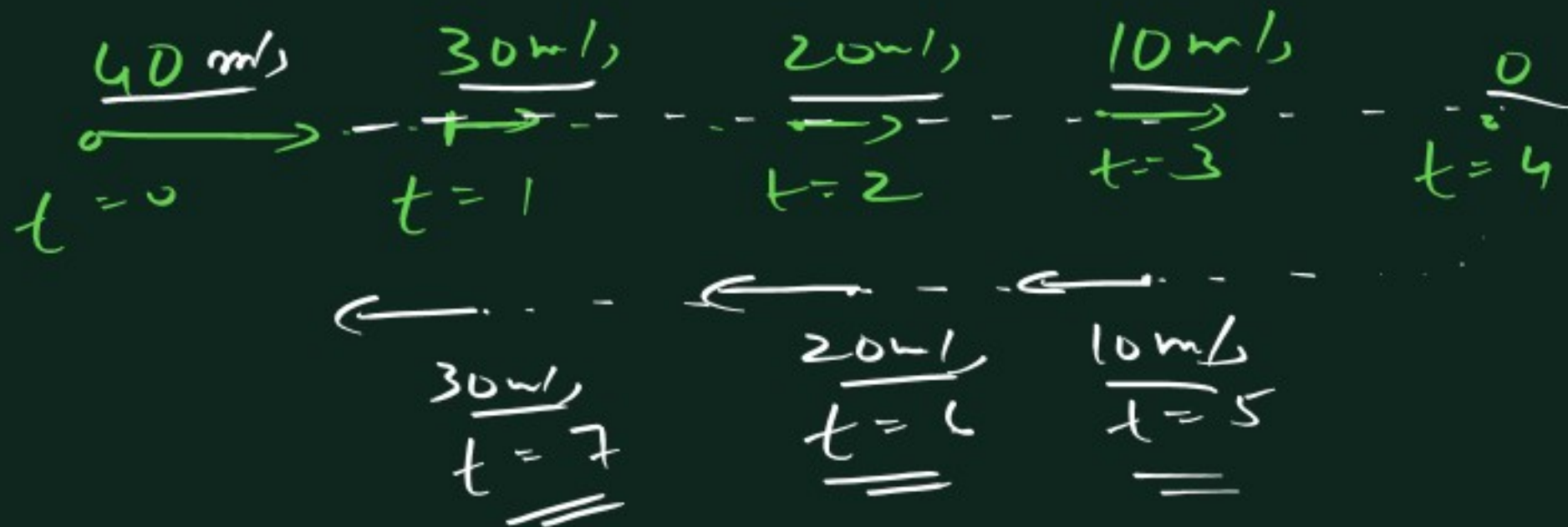
$$= 10 \text{ m/s}$$

$$t = 6 \text{ s}$$

$$v = 40 - 10 \times 6$$

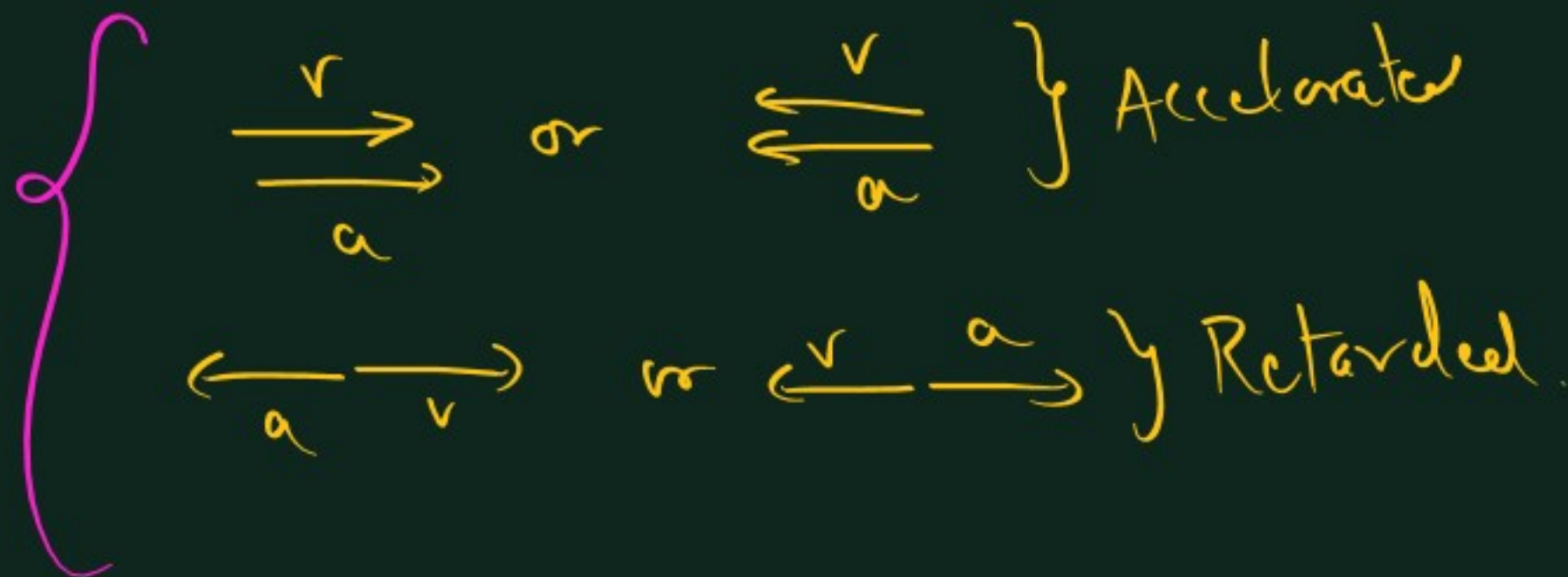
$$= -20$$

$$\text{Speed} = \underline{20 \text{ m/s}}$$



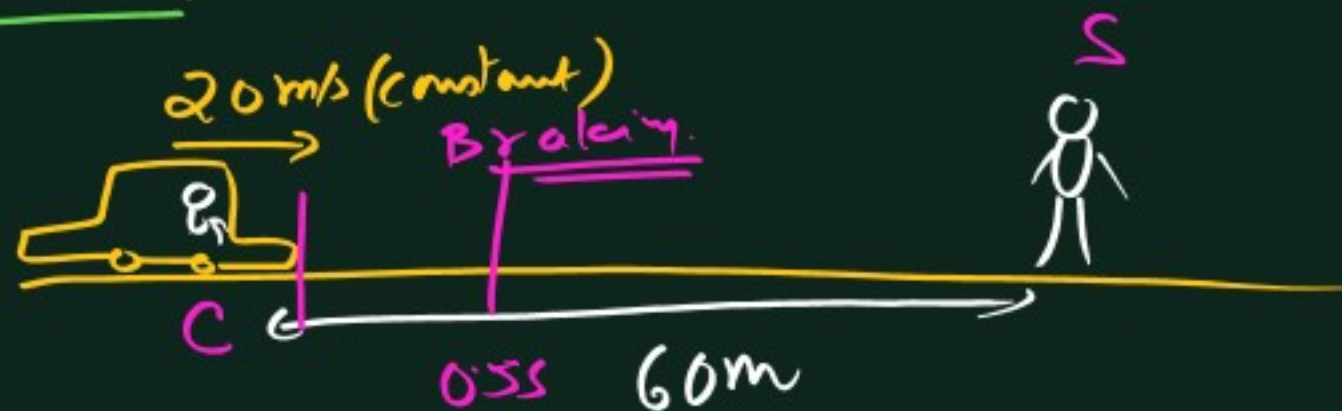
Retarded $\rightarrow t = 0$ to 4 s .

Accelerated $t = 4 \text{ s}$ & beyond

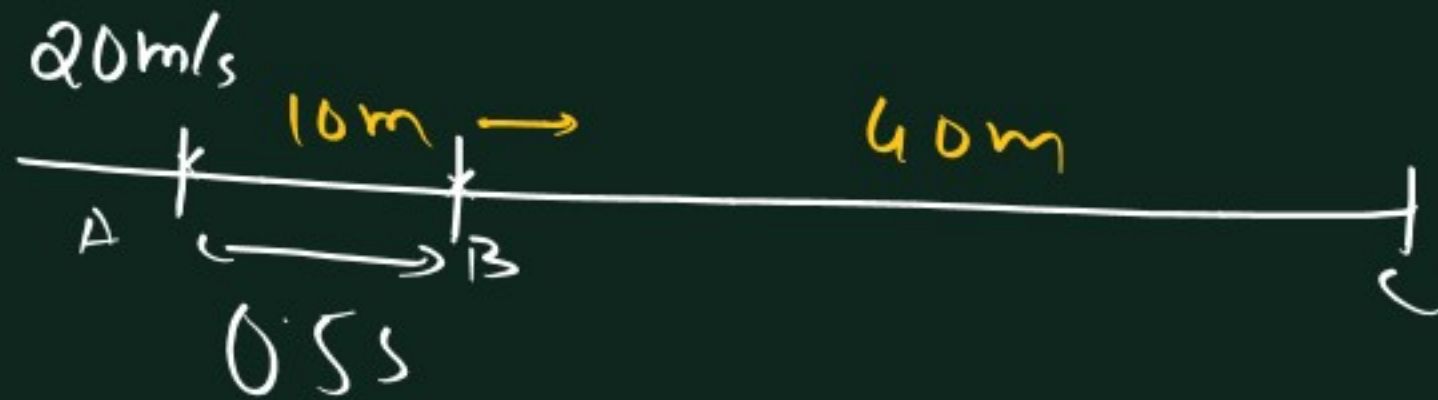


Reaction time

Q



Rxn time = time between seeing & applying brakes.



A to B $s_1 = v \times t = 20 \times 0.5$
 $= 10m$

From B

$$u = 20m/s$$

$$a = -5m/s^2$$

$$v = 0$$

$$v^2 = u^2 + 2as$$

$$\Rightarrow 0 = 20^2 - 2 \times 5 \times s$$

$$\Rightarrow s = \frac{400}{10} = \underline{\underline{40m}}$$

Total disp. of car = 10 + 40
= 50m

Person saved by 10m.

Q

$$\frac{1000}{3600} \rightarrow \frac{5}{18}$$

$$\frac{54 \times 5}{18} = \underline{15 \text{ m/s}}$$

$$u = 15$$

$$v = 0$$

$$\underline{t = 10 \text{ s.}}$$

$$\begin{aligned} S &= u_{\text{avg}} \times t \\ &= 7.5 \times 10 \\ &= \underline{\underline{75 \text{ m}}} \end{aligned}$$

Motion under gravity.



$$a = \frac{F}{m} = \frac{mg}{m} = g (\downarrow)$$

→ $g = 9.8 \text{ m/s}^2$
[Acceleration due to gravity]

(g never $\approx 9.8 \text{ m/s}^2$)

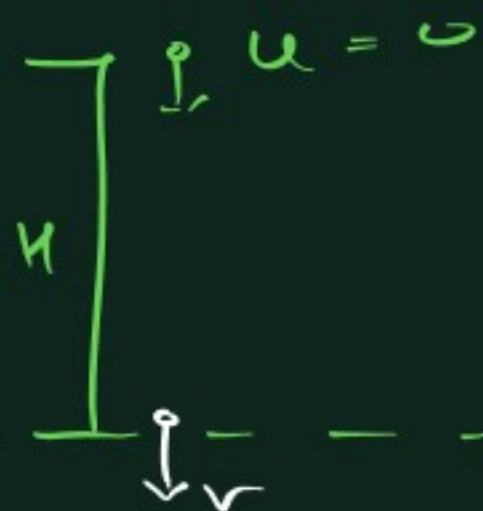
②



g



Case I



$\downarrow \oplus$ (Easier)

1) Find speed just before landing

2) time of fall

(air resistance = 0)

$$u = 0$$

$$a = +g(\downarrow) \text{ (or } +9.8)$$

$$s = +h(\downarrow)$$

$$v^2 = u^2 + 2as$$

$$\Rightarrow v^2 = 0 + 2gh$$

$$\Rightarrow \boxed{v = \sqrt{2gh}}$$

$$s = \cancel{ut} + \frac{1}{2}at^2$$

$$\Rightarrow h = \frac{1}{2}gt^2$$

$$\Rightarrow \boxed{t = \sqrt{\frac{2h}{g}}}$$

$$\sqrt{4.9} \neq 0.7$$

$$\sqrt{\frac{490}{100}} = \sqrt{\frac{490}{10}} = \frac{22.1}{10} = \textcircled{2.21}$$

$$\begin{array}{r} \underline{484} \\ 490 \end{array} \quad \begin{array}{r} \underline{529} \end{array}$$

$$\underline{2.2136}$$

$$\sqrt{5.7} = \sqrt{\frac{570}{10}}$$

$$(24 = 576) = \frac{23.9}{10} = \underline{\underline{2.39}}$$

$$(g = 10 \text{ m/s}^2)$$

$$\downarrow u = 0$$

$$a = +10$$

$$t = 1 \text{ s.}$$

$$s = ut + \frac{1}{2}at^2$$

$$= 0 + \frac{1}{2} \times 10 \times 1^2$$

$$= 5 \text{ m}$$

222

$$S_{2n} = u + \frac{a}{2}(2n-1)$$

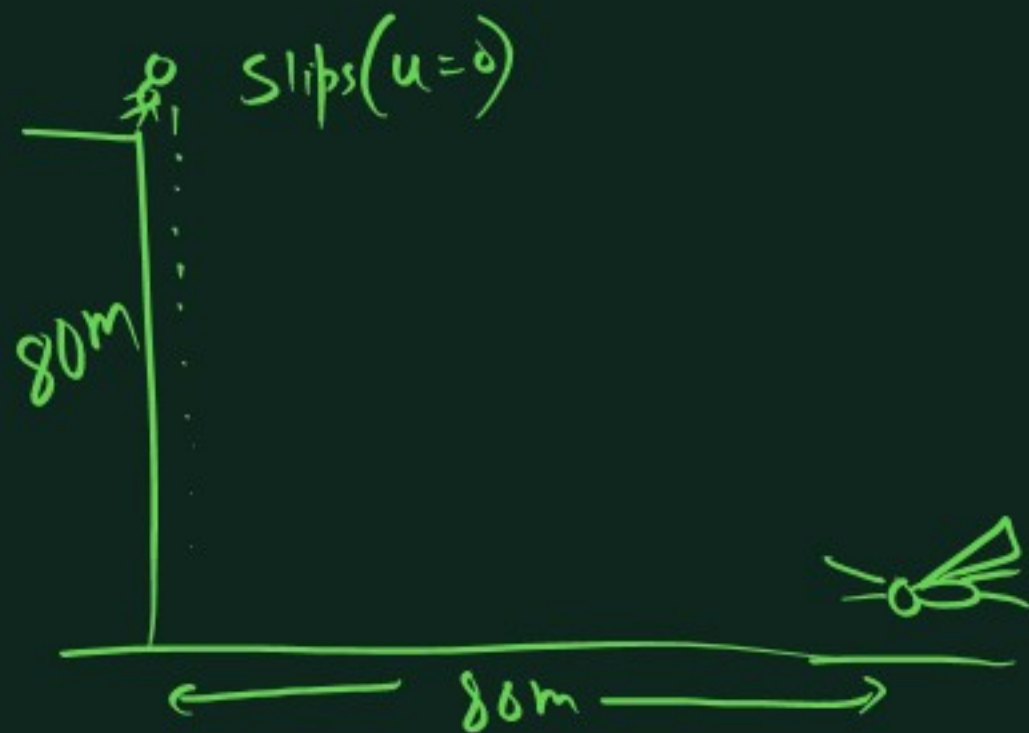
$$= 0 + 5(2 \times 1 - 1) = 15 \text{ m}$$

$$(u=0)$$

$$S_{1st} : S_{2nd} : S_{3rd} \dots = 1 : 3 : 5 \dots$$

$$= 5 \text{ m} : 15 \text{ m} : 25 \text{ m}$$

Q



$$g = 10 \text{ m/s}^2$$

Time to fall

$$t = \sqrt{\frac{2H}{g}}$$

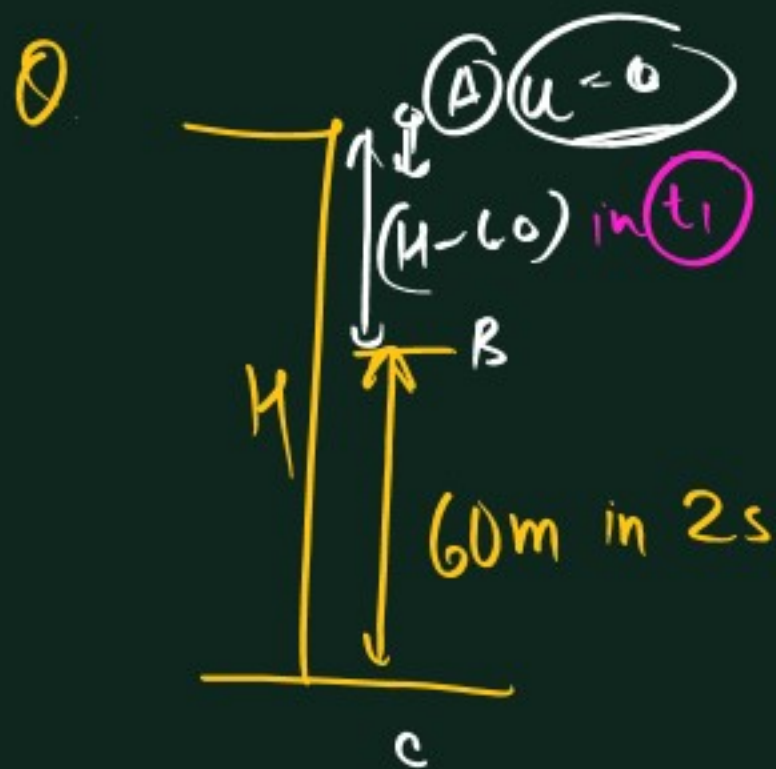
$$= \sqrt{\frac{2 \times 80}{10}} = 4$$

Speed here

$$s = v \times t$$

$$\Rightarrow v = \frac{s}{t}$$

$$= \frac{80}{4} = 20 \text{ m/s}$$



A to B

$$S = \frac{1}{2} at^2 \quad (u=0)$$

$$H-60 = \frac{1}{2} \times 10 \times t_1^2 \quad \text{--- (1)}$$

$$H-60 = 5t_1^2$$

A to C

$$S = \frac{1}{2} at^2$$

$$\Rightarrow H = \frac{1}{2} \times 10 \times (t_1+2)^2$$

$$\Rightarrow H = 5(t_1+2)^2 \quad \text{--- (2)}$$

put H in (1)

$$5(t_1+2)^2 - 60 = 5t_1^2$$

$$\Rightarrow 5(t_1^2 + 4 + 4t_1) - 60 = 5t_1^2$$

$$\Rightarrow 5t_1^2 + 20 + 20t_1 - 60 = 5t_1^2$$

$$\Rightarrow 20t_1 = 40$$

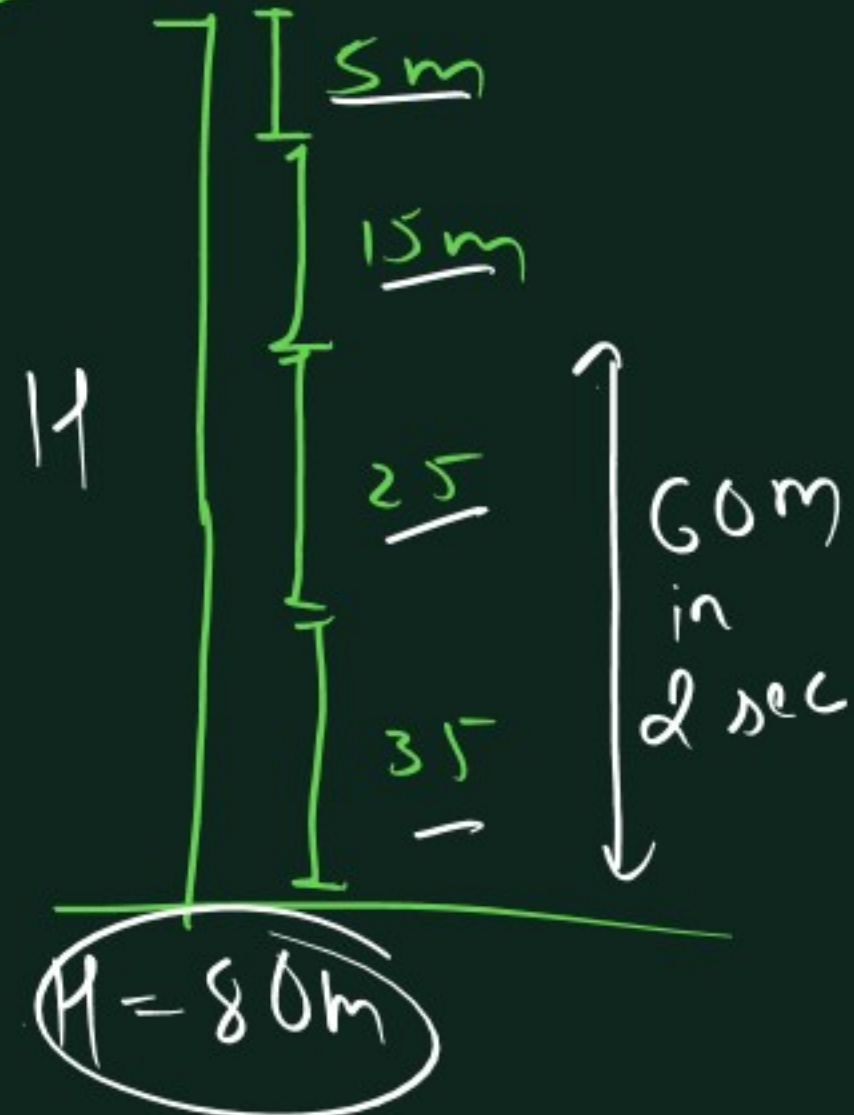
$$\Rightarrow \boxed{t_1 = 2\text{s}}$$

Put t_1 in (2)

$$H = 5(2+2)^2$$

$$= \underline{\underline{80\text{m}}}$$

2nd way



II



$$u = +u$$

$$a = -g$$

